14.1.6 Wrap-Up: Ticket Out the Door

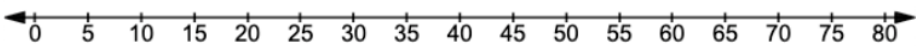
At Universal Studios, a person must be at least 48 inches (in.) tall to ride Revenge of the Mummy™.

1. Write an inequality representing the height, h, a person must be to go on the ride.

2. Circle all values from the list below that satisfy this inequality.

47.9 in.	100 in.	$\frac{48}{2}$ in.	48.5 in.
$70\frac{3}{4}$ in.	$48\frac{1}{4}$ in.	0 in.	400 in.
48.099 in.	48 in.	$50\frac{3}{4}$ in.	47.9 in.

3. Graph the inequality on the number line.



Unit 14, Lesson 2: Finding an Unknown Value in an Equation

Benchmark

MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.

Learning Targets

- ☐ I can determine an unknown decimal in a numeric equation with three numbers.
- ☐ I can determine an unknown fraction in a numeric equation with three numbers.

14.2.1 Warm-Up: What Is STEM?

Science, Technology, Engineering, and Mathematics (STEM) is a way of thinking and doing. STEM skills are useful to almost every career.

An individual does not need to be a master in all four areas of STEM to be successful; teamwork plays an important role. Playing to a person's strengths and working as a team creates a greater impact than working alone.

1. Describe a time where it was better for you to work as a team than alone.

2. List two jobs that involve working as a team but are in different fields of STEM.

14.2.2 Refresher: The Unknown Value

With your partner, complete the following.

In math, ***variables*** are used to represent unknown values.

In an algebraic expression, a ***variable*** is a letter that represents a number.

1. Circle the variable(s) in each expression.

$m + 5$ $\qquad$ $2x + y$ $\qquad$ $\frac{a}{4}$

Variables are found in algebraic equations.

An ***equation*** is a mathematical relation statement where two equivalent expressions or values are separated by an equal sign.

An ***algebraic equation*** is an equation with at least one variable.

2. Circle the algebraic equations.

$6 + 8 = 14$ $\qquad$ $5y = 35$ $\qquad$ $2w + 7 = 13$ $\qquad$ $\frac{12}{3} + 8 = 12$

To **solve** an equation means to find the solution.

The **solution(s)** to an algebraic equation is (are) the value(s) of the variable(s) that makes the equation true.

14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams

For some equations, the solution can be found mentally or by using reasoning.

1. The solution to the equation $n - 12.1 = 35.3$ can be found using reasoning.

   a. Discuss with your partner how to find the value of n in the equation $n - 12.1 = 35.3$.

   b. The equation is rewritten vertically with a box in the place of the variable. Write the solution in the box and check whether it makes the equation true.

$$\begin{array}{r} \boxed{} \\ \underline{-\ 12.1} \\ 35.3 \end{array}$$

Diagrams can also be helpful to find a solution.

2. Olivia gave equal amounts of money to each of her 3 children. Olivia gave each child $\$16.75$.

   The bar model represents the equation $\frac{m}{3} = 16.75$, where m is the total amount of money Olivia gave all of her children.

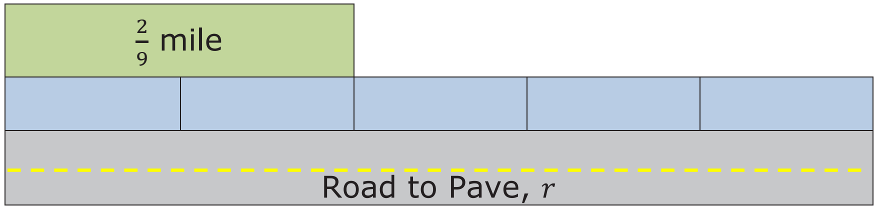
16.75	16.75	16.75
	m	

   a. How can the bar model be used to find the total amount of money Olivia gave all of her children, m?

   b. What number can be divided by 3 to get a result of 16.75?

$$\frac{\boxed{}}{3} = 16.75$$

3. A road construction crew paved $\frac{2}{9}$ mile, which is $\frac{2}{5}$ of a full paving job. The bar model represents the equation $\frac{2}{9} = \frac{2}{5}r$, where r is the length of the whole road being paved.



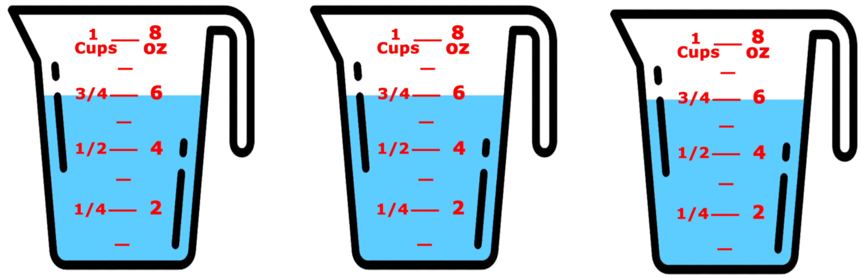
a. Discuss with your partner how the diagram represents the information given.

b. Describe how to use the bar model to solve the equation for r, the length of the road being paved.

c. How long is the road being paved?

d. Check your solution. $\frac{2}{9} = \frac{2}{5}\left(\boxed{\frac{}{}}\right)$

4. Donté used milk for 3 batches of a recipe. He used $\frac{3}{4}$ cup of milk for each batch.



The equation $m \div 3 = \frac{3}{4}$ can be used to find m, the amount of milk Donté used.

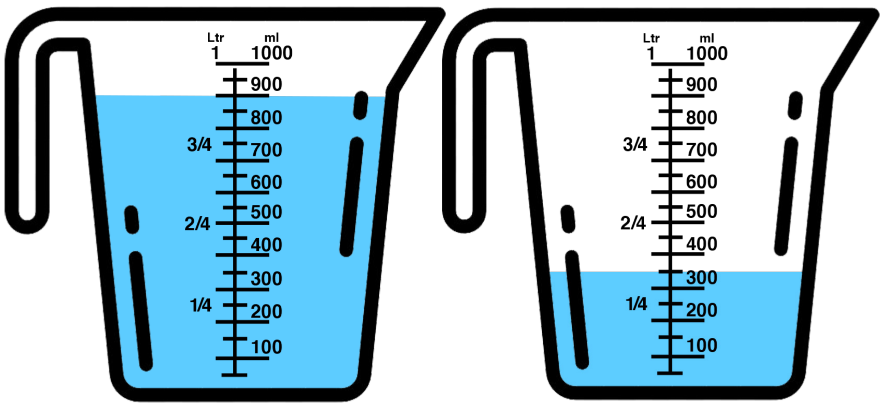
a. Use the measuring cups shown to solve the equation $m \div 3 = \frac{3}{4}$.

b. Check your solution. $\boxed{} \div 3 = \frac{3}{4}$

14.2.4 Collaboration: Solving Equations Using Reasoning & Diagrams

With your partner, solve the following problems using reasoning and diagrams.

1. Vincent had 0.9 liters (L) of water. He poured some into the mixing bowl and had 0.35 L left.



The bar model shown represents this situation.

$$\overbrace{\hspace{8em}}^{0.9}$$

w	0.35

a. Discuss with your partner how the bar model represents Vincent's situation. Then, write a brief summary of your discussion.

b. The equation $0.9 - w = 0.35$ can also be used to represent this situation. Find the value of w, the amount of water Vincent poured into the mixing bowl.

2. Josephine told her brother she would give him $\frac{1}{2}$ of the money they made from hosting a lemonade stand at their school carnival. She gave her brother $\$24.25$. This situation can be represented by the equation $\frac{1}{2}x = 24.25$ to find x, the amount of money they made at the lemonade stand.

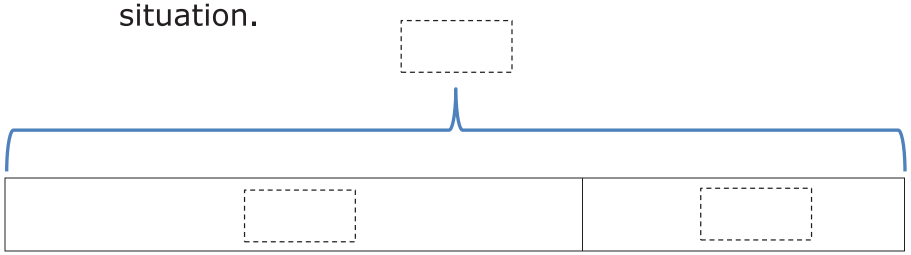
a. Use the bar model shown to represent the situation.

b. Find the value of x, the amount of money they made at the lemonade stand.

14.2.5 Your Turn!

1. DeAndré paid $\$58.54$ for a game at Store A, which was $\$9.21$ less than the price of the game at Store B. This situation can be represented by the equation $58.54 = b - 9.21$, where b is the price of the game at Store B.

    a. Complete the bar model diagram to represent this situation.



    b. Find the value of b.

2. Mr. Bing placed a new 5-gallon water bottle in the office water dispenser when he arrived in the morning. At the end of the day, the bottle had $2\frac{7}{8}$ gallons remaining. The equation $5 = x + 2\frac{7}{8}$ can be used to find x, the gallons of water dispensed from the bottle.

    a. Draw a diagram representing the situation.

    b. How much water was dispensed from the bottle? Include units.

14.2.6 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered…	
One mistake I learned from today…	One thing I am not sure about…YET!
One thing I am confident I know is…	

Unit 14, Lesson 3: Discovering Inverse Operations

Benchmark

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

Learning Targets

- ☐ I can use manipulatives to discover the role of inverse operations when solving one-step addition and subtraction equations.
- ☐ I can solve addition and subtraction one-step equations in one variable using manipulatives, drawings, and number lines to show the inverse operations.